ULRMS DATABASE ARCHITECTURE & SCHEMA SPECIFICATION

Unified Lab Record Management System • Comprehensive Catalog of Tables, Columns, Relationships & Constraints

GENERATED: Monday, November 9, 2026 • POSTGRESQL 16 (NEON ENGINE) • APP DESIGN DOCUMENT

TOTAL ACTIVE TABLES	TOTAL COLUMNS	CONSTRAINTS & RULES	FOREIGN RELATIONSHIPS
94 Tables	**1,128 Columns**	**817 Rules**	**220 Foreign Keys**
Relational Entities	Attributes & Fields	PK, FK, Unique & Checks	Relational Graph Bridges

ALPHABETICAL TABLE DIRECTORY INDEX (94 TABLES)

1. academic_years	(7 col)	2. activity_logs	(13 col)	3. admin_notes	(8 col)	4. assignment_files	(8 col)
5. assignment_targets	(13 col)	6. assignments	(40 col)	7. audit_logs	(11 col)	8. backup_codes	(7 col)
9. chat_messages	(7 col)	10. chat_sessions	(5 col)	11. class_enrollments	(6 col)	12. class_subjects	(5 col)
13. classes	(11 col)	14. coding_submissions	(9 col)	15. device_tests	(20 col)	16. document_folders	(8 col)
17. document_shares	(11 col)	18. document_view_logs	(5 col)	19. documents	(18 col)	20. equipment_shift_history	(9 col)
21. equipment_shift_requests	(12 col)	22. fee_categories	(8 col)	23. fee_payments	(11 col)	24. fee_structures	(11 col)
25. final_lab_marks	(19 col)	26. folder_shares	(11 col)	27. generated_reports	(6 col)	28. grade_history	(7 col)
29. grade_scale_history	(14 col)	30. grade_scales	(10 col)	31. grades	(25 col)	32. group_members	(5 col)
33. implementation_plans	(14 col)	34. import_history	(11 col)	35. item_maintenance_history	(10 col)	36. lab_attendance	(9 col)
37. lab_event_history	(12 col)	38. lab_items	(17 col)	39. lab_maintenance_history	(11 col)	40. lab_material_usage	(9 col)
41. lab_materials	(10 col)	42. labs	(14 col)	43. laptop_issuances	(20 col)	44. lecture_attendance	(7 col)
45. lecture_plans	(20 col)	46. lecture_poll_responses	(5 col)	47. lecture_polls	(7 col)	48. lecture_resources	(11 col)
49. lecture_sessions	(10 col)	50. meeting_participants	(11 col)	51. meeting_questions	(12 col)	52. meetings	(32 col)
53. notification_templates	(11 col)	54. notifications	(17 col)	55. procurement_committee	(6 col)	56. procurement_items	(17 col)
57. procurement_requests	(37 col)	58. query_logs	(12 col)	59. question_bank	(19 col)	60. quiz_assignments	(8 col)
61. quiz_submissions	(12 col)	62. quizzes	(15 col)	63. quotation_items	(7 col)	64. report_templates	(10 col)
65. school_calendar	(14 col)	66. schools	(20 col)	67. site_updates	(6 col)	68. student_fees	(11 col)
69. student_groups	(8 col)	70. student_training_progress	(10 col)	71. student_unit_mastery	(8 col)	72. subjects	(10 col)
73. submission_files	(8 col)	74. submission_revisions	(8 col)	75. submissions	(18 col)	76. system_settings	(5 col)
77. ticket_comments	(5 col)	78. ticket_issue_types	(7 col)	79. tickets	(17 col)	80. timetable_slots	(11 col)
81. timetables	(10 col)	82. training_exercises	(18 col)	83. training_modules	(15 col)	84. training_units	(8 col)
85. translations	(6 col)	86. user_sessions	(7 col)	87. users	(34 col)	88. vendor_quotations	(11 col)
89. vendors	(11 col)	90. whiteboard_files	(12 col)	91. whiteboard_participants	(7 col)	92. whiteboard_recording_shares	(9 col)
93. whiteboard_recordings	(14 col)	94. whiteboard_sessions	(17 col)				

TABLE: ACADEMIC_YEARS

7 Columns | PK: (id) | #1 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id CASCADE] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	year_label	character varying(20)	NOT NULL	-	-
4	start_date	date	NOT NULL	-	-
5	end_date	date	NOT NULL	-	-
6	is_current	boolean	NULL	false	-
7	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: ACTIVITY_LOGS

13 Columns | PK: (id) | #2 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id CASCADE] [UNIQUE]
2	user_id	uuid	NOT NULL	-	[FK -> users.id CASCADE]
3	activity_type	enum (ActivityType)	NULL	-	-
4	description	text	NOT NULL	-	-
5	description_hindi	text	NULL	-	-
6	entity_type	character varying(50)	NULL	-	-
7	entity_id	uuid	NULL	-	-
8	ip_address	character varying(50)	NULL	-	-
9	user_agent	text	NULL	-	-
10	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
11	school_id	uuid	NULL	-	-
12	action_type	text	NOT NULL	'system'	-
13	metadata	jsonb	NULL	-	-

Table Rules & Indexes: [Index activity_logs_created_at_idx]: USING btree (created_at) | [Index activity_logs_school_id_created_at_idx]: USING btree (school_id, created_at) | [Index activity_logs_user_id_idx]: USING btree (user_id)

TABLE: ADMIN_NOTES

8 Columns | PK: (id) | #3 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [UNIQUE]
2	title	character varying(255)	NOT NULL	-	-
3	content	text	NOT NULL	-	-
4	category	character varying(50)	NOT NULL	'general'	-
5	is_pinned	boolean	NOT NULL	false	-
6	author_id	uuid	NOT NULL	-	[FK -> users.id]
7	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
8	updated_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index admin_notes_author_id_idx]: USING btree (author_id) | [Index admin_notes_category_idx]: USING btree (category) | [Index admin_notes_is_pinned_idx]: USING btree (is_pinned)

TABLE: ASSIGNMENT_FILES

8 Columns | PK: (id) | #4 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> assignments.id CASCADE] [UNIQUE]
2	**assignment_id**	uuid	NOT NULL	-	[FK -> assignments.id CASCADE]
3	**file_name**	character varying(255)	NOT NULL	-	-
4	**file_type**	character varying(50)	NULL	-	-
5	**file_size**	integer	NULL	-	-
6	**file_url**	text	NOT NULL	-	-
7	**uploaded_by**	uuid	NOT NULL	-	-
8	**uploaded_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: ASSIGNMENT_TARGETS

13 Columns | PK: (id) | #5 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> assignments.id CASCADE] [FK -> student_groups.id] [UNIQUE]
2	**assignment_id**	uuid	NOT NULL	-	[FK -> assignments.id CASCADE]
3	**target_type**	enum (TargetType)	NOT NULL	-	-
4	**target_class_id**	uuid	NULL	-	-
5	**target_group_id**	uuid	NULL	-	[FK -> student_groups.id]
6	**target_student_id**	uuid	NULL	-	-
7	**assigned_by**	uuid	NOT NULL	-	[FK -> users.id]
8	**assigned_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
9	**special_instructions**	text	NULL	-	-
10	**extended_due_date**	timestamp without time zone	NULL	-	-
11	**publish_date**	timestamp without time zone	NULL	-	-
12	**due_date**	timestamp without time zone	NULL	-	-
13	**is_locked**	boolean	NOT NULL	false	-

TABLE: ASSIGNMENTS

40 Columns | PK: (id) | #6 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> users.id] [FK -> labs.id] [FK -> schools.id CASCADE] [FK -> subjects.id CASCADE] [FK -> training_modules.id] [UNIQUE]
2	**school_id**	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	**subject_id**	uuid	NOT NULL	-	[FK -> subjects.id CASCADE]
4	**lab_id**	uuid	NULL	-	[FK -> labs.id]
5	**created_by**	uuid	NOT NULL	-	[FK -> users.id]
6	**title**	character varying(255)	NOT NULL	-	-

TABLE: ASSIGNMENTS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
7	title_hindi	character varying(255)	NULL	-	-
8	title_regional	character varying(255)	NULL	-	-
9	description	text	NULL	-	-
10	description_hindi	text	NULL	-	-
11	description_regional	text	NULL	-	-
12	experiment_number	character varying(20)	NULL	-	-
13	assignment_type	enum (AssignmentType)	NOT NULL	-	-
14	programming_language	character varying(50)	NULL	-	-
15	aim	text	NULL	-	-
16	aim_hindi	text	NULL	-	-
17	theory	text	NULL	-	-
18	theory_hindi	text	NULL	-	-
19	procedure	text	NULL	-	-
20	procedure_hindi	text	NULL	-	-
21	expected_output	text	NULL	-	-
22	reference_code	text	NULL	-	-
23	attachments	jsonb	NULL	-	-
24	max_marks	integer	NULL	100	-
25	passing_marks	integer	NULL	35	-
26	viva_marks	integer	NULL	20	-
27	practical_marks	integer	NULL	60	-
28	output_marks	integer	NULL	20	-
29	publish_date	timestamp without time zone	NULL	-	-
30	due_date	timestamp without time zone	NULL	-	-
31	late_submission_allowed	boolean	NULL	true	-
32	late_penalty_percent	integer	NULL	10	-
33	status	enum (AssignmentStatus)	NULL	'draft'::"AssignmentStatus"	-
34	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
35	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
36	academic_year_id	uuid	NULL	-	[FK -> academic_years.id]
37	pdf_attachment_url	text	NULL	-	-
38	pdf_attachment_name	character varying(255)	NULL	-	-
39	pdf_cloudinary_id	character varying(255)	NULL	-	-
40	training_module_id	uuid	NULL	-	[FK -> training_modules.id]

Table Rules & Indexes: [Index assignments_academic_year_id_idx]: USING btree (academic_year_id) | [Index assignments_school_id_academic_year_id_idx]: USING btree (school_id, academic_year_id) | [Index assignments_status_idx]: USING btree (status)

TABLE: AUDIT_LOGS

11 Columns | PK: (id) | #7 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id] [FK -> users.id] [UNIQUE]
2	**user_id**	uuid	NULL	-	[FK -> users.id]
3	**school_id**	uuid	NULL	-	[FK -> schools.id]
4	**action**	character varying(50)	NOT NULL	-	-
5	**entity_type**	character varying(50)	NOT NULL	-	-
6	**entity_id**	uuid	NULL	-	-
7	**entity_name**	character varying(255)	NULL	-	-
8	**details**	jsonb	NULL	-	-
9	**ip_address**	character varying(45)	NULL	-	-
10	**user_agent**	text	NULL	-	-
11	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_audit_logs_action]: USING btree (action) | [Index idx_audit_logs_created]: USING btree (created_at DESC) | [Index idx_audit_logs_entity]: USING btree (entity_type, entity_id)

TABLE: BACKUP_CODES

7 Columns | PK: (id) | #8 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id CASCADE] [UNIQUE]
2	**user_id**	uuid	NOT NULL	-	[FK -> users.id CASCADE]
3	**code_hash**	character varying(255)	NOT NULL	-	-
4	**turn_index**	integer	NOT NULL	-	-
5	**is_used**	boolean	NOT NULL	false	-
6	**used_at**	timestamp without time zone	NULL	-	-
7	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_backup_codes_code_hash]: USING btree (code_hash) | [Index idx_backup_codes_user_id]: USING btree (user_id) | [Index idx_backup_codes_user_unused]: USING btree (user_id, is_used)

TABLE: CHAT_MESSAGES

7 Columns | PK: (id) | #9 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> chat_sessions.id] [UNIQUE]
2	**session_id**	uuid	NOT NULL	-	[FK -> chat_sessions.id]
3	**role**	character varying(50)	NOT NULL	-	-
4	**content**	text	NOT NULL	-	-
5	**sql**	text	NULL	-	-
6	**provider**	character varying(50)	NULL	-	-
7	**created_at**	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

TABLE: CHAT_SESSIONS

5 Columns | PK: (id) | #10 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [UNIQUE]
2	user_id	uuid	NOT NULL	-	[FK -> users.id]
3	title	character varying(255)	NOT NULL	-	-
4	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
5	updated_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

TABLE: CLASS_ENROLLMENTS

6 Columns | PK: (id) | #11 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> classes.id CASCADE] [FK -> users.id CASCADE] [UNIQUE]
2	student_id	uuid	NOT NULL	-	[FK -> users.id CASCADE]
3	class_id	uuid	NOT NULL	-	[FK -> classes.id CASCADE]
4	roll_number	integer	NULL	-	-
5	enrollment_date	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
6	status	enum (EnrollmentStatus)	NULL	'active'::"EnrollmentStatus"	-

Table Rules & Indexes: [Index class_enrollments_student_id_class_id_key]: USING btree (student_id, class_id) | [Index idx_enrollments_class]: USING btree (class_id) | [Index idx_enrollments_student]: USING btree (student_id)

TABLE: CLASS_SUBJECTS

5 Columns | PK: (id) | #12 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> classes.id CASCADE] [FK -> users.id] [FK -> users.id] [FK -> subjects.id CASCADE] [UNIQUE]
2	class_id	uuid	NOT NULL	-	[FK -> classes.id CASCADE]
3	subject_id	uuid	NOT NULL	-	[FK -> subjects.id CASCADE]
4	instructor_id	uuid	NULL	-	[FK -> users.id]
5	lab_instructor_id	uuid	NULL	-	[FK -> users.id]

Table Rules & Indexes: [Index class_subjects_class_id_subject_id_key]: USING btree (class_id, subject_id)

TABLE: CLASSES

11 Columns | PK: (id) | #13 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id CASCADE] [FK -> users.id] [FK -> schools.id CASCADE] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	academic_year_id	uuid	NOT NULL	-	[FK -> academic_years.id CASCADE]
4	name	character varying(100)	NOT NULL	-	-
5	name_hindi	character varying(100)	NULL	-	-
6	grade_level	integer	NOT NULL	-	-
7	section	character varying(10)	NULL	-	-
8	stream	character varying(50)	NULL	-	-

TABLE: CLASSES (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
9	class_teacher_id	uuid	NULL	-	[FK -> users.id]
10	max_students	integer	NULL	60	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index classes_academic_year_id_idx]: USING btree (academic_year_id) | [Index classes_school_id_academic_year_id_idx]: USING btree (school_id, academic_year_id) | [Index classes_school_id_name_academic_year_id_key]: USING btree (school_id, name, academic_year_id)

TABLE: CODING_SUBMISSIONS

9 Columns | PK: (id) | #14 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> training_exercises.id] [FK -> users.id] [UNIQUE]
2	exercise_id	uuid	NOT NULL	-	[FK -> training_exercises.id]
3	student_id	uuid	NOT NULL	-	[FK -> users.id]
4	code	text	NOT NULL	-	-
5	status	text	NOT NULL	-	-
6	output	text	NULL	-	-
7	test_results	jsonb	NULL	-	-
8	ai_socratic_review	text	NULL	-	-
9	submitted_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

TABLE: DEVICE_TESTS

20 Columns | PK: (id) | #15 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id CASCADE] [UNIQUE]
2	user_id	uuid	NOT NULL	-	[FK -> users.id CASCADE] [UNIQUE]
3	camera_status	character varying(50)	NULL	-	-
4	camera_tested_at	timestamp without time zone	NULL	-	-
5	camera_device_id	text	NULL	-	-
6	camera_device_name	text	NULL	-	-
7	mic_status	character varying(50)	NULL	-	-
8	mic_tested_at	timestamp without time zone	NULL	-	-
9	mic_device_id	text	NULL	-	-
10	mic_device_name	text	NULL	-	-
11	speaker_status	character varying(50)	NULL	-	-
12	speaker_tested_at	timestamp without time zone	NULL	-	-
13	speaker_device_id	text	NULL	-	-
14	speaker_device_name	text	NULL	-	-
15	speaker_volume	integer	NULL	-	-
16	user_agent	text	NULL	-	-

TABLE: DEVICE_TESTS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
17	platform	character varying(50)	NULL	-	-
18	browser	character varying(100)	NULL	-	-
19	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
20	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index device_tests_user_id_key]: USING btree (user_id) | [Index idx_device_tests_user_id]: USING btree (user_id)

TABLE: DOCUMENT_FOLDERS

8 Columns | PK: (id) | #16 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> document_folders.id SET NULL] [FK -> schools.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]
3	parent_id	uuid	NULL	-	[FK -> document_folders.id SET NULL]
4	name	character varying(255)	NOT NULL	-	-
5	created_by	uuid	NOT NULL	-	[FK -> users.id]
6	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
7	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
8	deleted_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Index idx_document_folders_parent_id]: USING btree (parent_id) | [Index idx_document_folders_school_id]: USING btree (school_id)

TABLE: DOCUMENT_SHARES

11 Columns | PK: (id) | #17 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> documents.id CASCADE] [FK -> users.id] [FK -> classes.id CASCADE] [FK -> student_groups.id CASCADE] [FK -> users.id CASCADE] [UNIQUE] [CHECK]
2	document_id	uuid	NOT NULL	-	[FK -> documents.id CASCADE]
3	shared_by_id	uuid	NOT NULL	-	[FK -> users.id]
4	target_type	enum (document_share_target_type)	NOT NULL	-	[CHECK]
5	target_class_id	uuid	NULL	-	[FK -> classes.id CASCADE] [CHECK]
6	target_group_id	uuid	NULL	-	[FK -> student_groups.id CASCADE] [CHECK]
7	target_user_id	uuid	NULL	-	[FK -> users.id CASCADE] [CHECK]
8	message	text	NULL	-	-
9	permission	enum (share_permission)	NOT NULL	'download'	-
10	shared_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
11	expires_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Composite valid_target]: CHECK ((((target_type = 'class'::document_share_target_type) AND (target_class_id IS NOT NULL)) OR ((target_type = 'group'::document_share_target_type) AND (target_group_id IS NOT NULL)) OR ((target_type = ANY (ARRAY['instructor'::document_share_target_type, 'admin'::document_share_target_type, 'student'::document_share_target_type])) AND (target_user_id IS NOT NULL)))) | [Index idx_document_shares_class]: USING btree (target_class_id) | [Index idx_document_shares_document]: USING btree (document_id) | [Index idx_document_shares_group]: USING btree (target_group_id)

TABLE: DOCUMENT_VIEW_LOGS

5 Columns | PK: (id) | #18 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> documents.id] [FK -> users.id] [UNIQUE]
2	document_id	uuid	NOT NULL	-	[FK -> documents.id]
3	user_id	uuid	NOT NULL	-	[FK -> users.id]
4	viewed_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
5	ip_address	character varying(50)	NULL	-	-

Table Rules & Indexes: [Index document_view_logs_document_id_idx]: USING btree (document_id) | [Index document_view_logs_user_id_idx]: USING btree (user_id)

TABLE: DOCUMENTS

18 Columns | PK: (id) | #19 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> document_folders.id SET NULL] [FK -> schools.id] [FK -> users.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]
3	uploaded_by	uuid	NOT NULL	-	[FK -> users.id]
4	folder_id	uuid	NULL	-	[FK -> document_folders.id SET NULL]
5	name	character varying(255)	NOT NULL	-	-
6	description	text	NULL	-	-
7	file_name	character varying(255)	NOT NULL	-	-
8	file_type	character varying(50)	NOT NULL	-	-
9	mime_type	character varying(100)	NOT NULL	-	-
10	file_size	integer	NOT NULL	-	-
11	cloudinary_id	character varying(255)	NOT NULL	-	-
12	url	text	NOT NULL	-	-
13	is_public	boolean	NULL	false	-
14	category	character varying(100)	NULL	-	-
15	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
16	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
17	deleted_at	timestamp without time zone	NULL	-	-
18	deleted_by	uuid	NULL	-	[FK -> users.id]

Table Rules & Indexes: [Index idx_documents_deleted_at]: USING btree (deleted_at) | [Index idx_documents_folder_id]: USING btree (folder_id) | [Index idx_documents_school_id]: USING btree (school_id)

TABLE: EQUIPMENT_SHIFT_HISTORY

9 Columns | PK: (id) | #20 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> labs.id] [FK -> lab_items.id CASCADE] [FK -> equipment_shift_requests.id] [FK -> users.id] [FK -> labs.id] [UNIQUE]
2	item_id	uuid	NOT NULL	-	[FK -> lab_items.id CASCADE]
3	from_lab_id	uuid	NOT NULL	-	[FK -> labs.id]
4	to_lab_id	uuid	NOT NULL	-	[FK -> labs.id]

TABLE: EQUIPMENT_SHIFT_HISTORY (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
5	shifted_by	uuid	NOT NULL	-	[FK -> users.id]
6	approved_by	uuid	NULL	-	[FK -> users.id]
7	shift_request_id	uuid	NULL	-	[FK -> equipment_shift_requests.id]
8	notes	text	NULL	-	-
9	shifted_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_shift_history_item]: USING btree (item_id)

TABLE: EQUIPMENT_SHIFT_REQUESTS

12 Columns | PK: (id) | #21 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> labs.id] [FK -> lab_items.id CASCADE] [FK -> users.id] [FK -> labs.id] [UNIQUE]
2	item_id	uuid	NOT NULL	-	[FK -> lab_items.id CASCADE]
3	from_lab_id	uuid	NOT NULL	-	[FK -> labs.id]
4	to_lab_id	uuid	NOT NULL	-	[FK -> labs.id]
5	requested_by	uuid	NOT NULL	-	[FK -> users.id]
6	approved_by	uuid	NULL	-	[FK -> users.id]
7	status	enum (equipment_shift_status)	NULL	'pending'	-
8	reason	text	NULL	-	-
9	admin_notes	text	NULL	-	-
10	requested_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
11	approved_at	timestamp without time zone	NULL	-	-
12	completed_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Index idx_shift_requests_item]: USING btree (item_id) | [Index idx_shift_requests_status]: USING btree (status)

TABLE: FEE_CATEGORIES

8 Columns | PK: (id) | #22 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id CASCADE] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	name	character varying(100)	NOT NULL	-	-
4	name_hindi	character varying(100)	NULL	-	-
5	description	text	NULL	-	-
6	is_recurring	boolean	NULL	true	-
7	frequency	enum (FeeFrequency)	NULL	'yearly'::"FeeFrequency"	-
8	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: FEE_PAYMENTS

11 Columns | PK: (id) | #23 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> student_fees.id] [UNIQUE]
2	student_fee_id	uuid	NOT NULL	-	[FK -> student_fees.id]
3	student_id	uuid	NOT NULL	-	-
4	amount	numeric	NOT NULL	-	-
5	payment_mode	enum (PaymentMode)	NOT NULL	-	-
6	transaction_id	character varying(100)	NULL	-	-
7	receipt_number	character varying(50)	NULL	-	[UNIQUE]
8	payment_date	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
9	collected_by	uuid	NOT NULL	-	[FK -> users.id]
10	remarks	text	NULL	-	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index fee_payments_receipt_number_key]: USING btree (receipt_number)

TABLE: FEE_STRUCTURES

11 Columns | PK: (id) | #24 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> classes.id] [FK -> fee_categories.id] [FK -> schools.id CASCADE] [FK -> subjects.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	academic_year_id	uuid	NOT NULL	-	[FK -> academic_years.id]
4	fee_category_id	uuid	NOT NULL	-	[FK -> fee_categories.id]
5	class_id	uuid	NULL	-	[FK -> classes.id]
6	subject_id	uuid	NULL	-	[FK -> subjects.id]
7	amount	numeric	NOT NULL	-	-
8	currency	character varying(10)	NULL	'INR'	-
9	due_date	timestamp without time zone	NULL	-	-
10	concession_applicable	boolean	NULL	true	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: FINAL_LAB_MARKS

19 Columns | PK: (id) | #25 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> classes.id] [FK -> users.id] [FK -> users.id] [FK -> subjects.id] [UNIQUE]
2	student_id	uuid	NOT NULL	-	[FK -> users.id]
3	subject_id	uuid	NOT NULL	-	[FK -> subjects.id]
4	class_id	uuid	NOT NULL	-	[FK -> classes.id]
5	academic_year_id	uuid	NOT NULL	-	[FK -> academic_years.id]
6	total_assignments	integer	NOT NULL	-	-

TABLE: FINAL_LAB_MARKS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
7	completed_assignments	integer	NOT NULL	-	-
8	internal_marks	numeric	NULL	-	-
9	viva_average	numeric	NULL	-	-
10	practical_exam_marks	numeric	NULL	-	-
11	total_marks	numeric	NULL	-	-
12	max_marks	numeric	NULL	-	-
13	percentage	numeric	NULL	-	-
14	grade_letter	character varying(5)	NULL	-	-
15	grade_points	numeric	NULL	-	-
16	remarks	text	NULL	-	-
17	is_pass	boolean	NULL	-	-
18	finalized_by	uuid	NULL	-	[FK -> users.id]
19	finalized_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Index final_lab_marks_student_id_subject_id_academic_year_id_key]: USING btree (student_id, subject_id, academic_year_id)

TABLE: FOLDER_SHARES

11 Columns | PK: (id) | #26 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> document_folders.id CASCADE] [FK -> users.id] [FK -> classes.id CASCADE] [FK -> student_groups.id CASCADE] [FK -> users.id CASCADE] [UNIQUE]
2	folder_id	uuid	NOT NULL	-	[FK -> document_folders.id CASCADE]
3	shared_by_id	uuid	NOT NULL	-	[FK -> users.id]
4	target_type	enum (document_share_target_type)	NOT NULL	-	-
5	target_class_id	uuid	NULL	-	[FK -> classes.id CASCADE]
6	target_group_id	uuid	NULL	-	[FK -> student_groups.id CASCADE]
7	target_user_id	uuid	NULL	-	[FK -> users.id CASCADE]
8	message	text	NULL	-	-
9	permission	enum (share_permission)	NOT NULL	'download'	-
10	shared_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
11	expires_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Index idx_folder_shares_class]: USING btree (target_class_id) | [Index idx_folder_shares_folder]: USING btree (folder_id) | [Index idx_folder_shares_group]: USING btree (target_group_id)

TABLE: GENERATED_REPORTS

6 Columns | PK: (id) | #27 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> report_templates.id] [UNIQUE]
2	template_id	uuid	NOT NULL	-	[FK -> report_templates.id]
3	generated_by	uuid	NOT NULL	-	[FK -> users.id]

TABLE: GENERATED_REPORTS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
4	file_url	text	NOT NULL	-	-
5	parameters	jsonb	NULL	-	-
6	generated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: GRADE_HISTORY

7 Columns | PK: (id) | #28 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> grades.id CASCADE] [UNIQUE]
2	grade_id	uuid	NOT NULL	-	[FK -> grades.id CASCADE]
3	previous_marks	jsonb	NOT NULL	-	-
4	new_marks	jsonb	NOT NULL	-	-
5	reason	text	NULL	-	-
6	modified_by	uuid	NOT NULL	-	-
7	modified_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: GRADE_SCALE_HISTORY

14 Columns | PK: (id) | #29 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> grade_scales.id CASCADE] [UNIQUE]
2	grade_scale_id	uuid	NOT NULL	-	[FK -> grade_scales.id CASCADE]
3	action	character varying(50)	NOT NULL	-	-
4	previous_letter	character varying(10)	NULL	-	-
5	previous_min_pct	integer	NULL	-	-
6	previous_max_pct	integer	NULL	-	-
7	previous_points	numeric	NULL	-	-
8	new_letter	character varying(10)	NULL	-	-
9	new_min_pct	integer	NULL	-	-
10	new_max_pct	integer	NULL	-	-
11	new_points	numeric	NULL	-	-
12	changed_by_id	uuid	NOT NULL	-	[FK -> users.id]
13	changed_at	timestamp with time zone	NULL	CURRENT_TIMESTAMP	-
14	reason	text	NULL	-	-

Table Rules & Indexes: [Index idx_grade_scale_history_changed_at]: USING btree (changed_at) | [Index idx_grade_scale_history_scale_id]: USING btree (grade_scale_id)

TABLE: GRADE_SCALES

10 Columns | PK: (id) | #30 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id] [UNIQUE]

TABLE: GRADE_SCALES (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]
3	grade_letter	character varying(5)	NOT NULL	-	-
4	grade_point	numeric	NOT NULL	-	-
5	min_percentage	integer	NOT NULL	-	-
6	max_percentage	integer	NOT NULL	-	-
7	description	text	NULL	-	-
8	is_active	boolean	NOT NULL	true	-
9	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
10	updated_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index grade_scales_school_id_grade_letter_key]: USING btree (school_id, grade_letter)

TABLE: GRADES

25 Columns | PK: (id) | #31 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> users.id] [FK -> users.id] [FK -> submissions.id CASCADE] [UNIQUE]
2	submission_id	uuid	NOT NULL	-	[FK -> submissions.id CASCADE] [UNIQUE]
3	student_id	uuid	NOT NULL	-	-
4	graded_by	uuid	NOT NULL	-	[FK -> users.id]
5	practical_marks	numeric	NULL	0	-
6	output_marks	numeric	NULL	0	-
7	viva_marks	numeric	NULL	0	-
8	total_marks	numeric	NULL	-	-
9	max_marks	numeric	NULL	-	-
10	percentage	numeric	NULL	-	-
11	grade_letter	character varying(5)	NULL	-	-
12	late_penalty_marks	numeric	NULL	0	-
13	final_marks	numeric	NULL	-	-
14	code_feedback	text	NULL	-	-
15	code_feedback_hindi	text	NULL	-	-
16	output_feedback	text	NULL	-	-
17	general_remarks	text	NULL	-	-
18	general_remarks_hindi	text	NULL	-	-
19	is_published	boolean	NULL	false	-
20	published_at	timestamp without time zone	NULL	-	-
21	graded_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
22	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
23	academic_year_id	uuid	NULL	-	[FK -> academic_years.id]

TABLE: GRADES (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
24	modified_at	timestamp without time zone	NULL	-	-
25	modified_by_id	uuid	NULL	-	[FK -> users.id]

Table Rules & Indexes: [Index grades_academic_year_id_idx]: USING btree (academic_year_id) | [Index grades_student_id_idx]: USING btree (student_id) | [Index grades_submission_id_key]: USING btree (submission_id)

TABLE: GROUP_MEMBERS

5 Columns | PK: (id) | #32 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> student_groups.id CASCADE] [FK -> users.id CASCADE] [UNIQUE]
2	group_id	uuid	NOT NULL	-	[FK -> student_groups.id CASCADE]
3	student_id	uuid	NOT NULL	-	[FK -> users.id CASCADE]
4	role	enum (GroupRole)	NULL	'member'::"GroupRole"	-
5	joined_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index group_members_group_id_student_id_key]: USING btree (group_id, student_id)

TABLE: IMPLEMENTATION_PLANS

14 Columns | PK: (id) | #33 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	school_id	uuid	NULL	-	-
3	created_by_id	uuid	NULL	-	-
4	title	character varying(255)	NOT NULL	-	-
5	description	text	NULL	-	-
6	category	character varying(100)	NULL	'General'	-
7	status	character varying(50)	NOT NULL	'draft'	-
8	started_at	timestamp without time zone	NULL	-	-
9	ended_at	timestamp without time zone	NULL	-	-
10	tasks	jsonb	NULL	'[]'	-
11	outcomes	text	NULL	-	-
12	metadata	jsonb	NULL	'{}'	-
13	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
14	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_implementation_plans_category]: USING btree (category) | [Index idx_implementation_plans_dates]: USING btree (started_at, ended_at) | [Index idx_implementation_plans_status]: USING btree (status)

TABLE: IMPORT_HISTORY

11 Columns | PK: (id) | #34 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> labs.id CASCADE] [FK -> schools.id] [FK -> users.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]

TABLE: IMPORT_HISTORY (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
3	lab_id	uuid	NOT NULL	-	[FK -> labs.id CASCADE]
4	uploaded_by	uuid	NOT NULL	-	[FK -> users.id]
5	file_name	character varying(255)	NOT NULL	-	-
6	file_size	integer	NULL	-	-
7	items_imported	integer	NOT NULL	-	-
8	items_failed	integer	NULL	0	-
9	status	character varying(50)	NULL	'completed'	-
10	errors	jsonb	NULL	-	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_import_history_created_at]: USING btree (created_at DESC) | [Index idx_import_history_lab_id]: USING btree (lab_id) | [Index idx_import_history_school_id]: USING btree (school_id)

TABLE: ITEM_MAINTENANCE_HISTORY

10 Columns | PK: (id) | #35 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lab_items.id CASCADE] [FK -> users.id] [UNIQUE]
2	item_id	uuid	NOT NULL	-	[FK -> lab_items.id CASCADE]
3	recorded_by	uuid	NOT NULL	-	[FK -> users.id]
4	type	character varying(50)	NOT NULL	-	-
5	description	text	NOT NULL	-	-
6	cost	numeric	NULL	-	-
7	vendor	character varying(255)	NULL	-	-
8	part_name	character varying(255)	NULL	-	-
9	resolved_at	timestamp without time zone	NULL	-	-
10	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: LAB_ATTENDANCE

9 Columns | PK: (id) | #36 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> classes.id] [FK -> labs.id CASCADE] [FK -> users.id] [FK -> users.id] [UNIQUE]
2	lab_id	uuid	NOT NULL	-	[FK -> labs.id CASCADE]
3	class_id	uuid	NOT NULL	-	[FK -> classes.id]
4	student_id	uuid	NOT NULL	-	[FK -> users.id]
5	date	date	NOT NULL	-	-
6	status	enum (AttendanceStatus)	NULL	'present'::"AttendanceSta...	-
7	marked_by	uuid	NOT NULL	-	[FK -> users.id]
8	remarks	text	NULL	-	-
9	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: LAB_EVENT_HISTORY

12 Columns | PK: (id) | #37 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lab_items.id] [FK -> labs.id] [FK -> users.id] [FK -> users.id] [FK -> users.id] [FK -> procurement_items.id] [FK -> procurement_requests.id] [UNIQUE]
2	lab_id	uuid	NOT NULL	-	[FK -> labs.id]
3	event_type	character varying(50)	NOT NULL	-	-
4	description	text	NULL	-	-
5	item_id	uuid	NULL	-	[FK -> lab_items.id]
6	item_details	jsonb	NULL	-	-
7	old_incharge_id	uuid	NULL	-	[FK -> users.id]
8	new_incharge_id	uuid	NULL	-	[FK -> users.id]
9	procurement_request_id	uuid	NULL	-	[FK -> procurement_requests.id]
10	procurement_item_id	uuid	NULL	-	[FK -> procurement_items.id]
11	performed_by_id	uuid	NOT NULL	-	[FK -> users.id]
12	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_lab_event_history_created_at]: USING btree (created_at DESC) | [Index idx_lab_event_history_event_type]: USING btree (event_type) | [Index idx_lab_event_history_lab_id]: USING btree (lab_id)

TABLE: LAB_ITEMS

17 Columns | PK: (id) | #38 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> labs.id CASCADE] [FK -> schools.id] [UNIQUE]
2	lab_id	uuid	NULL	-	[FK -> labs.id CASCADE]
3	school_id	uuid	NULL	-	[FK -> schools.id]
4	item_type	character varying(50)	NOT NULL	-	-
5	item_number	character varying(100)	NOT NULL	-	-
6	brand	character varying(100)	NULL	-	-
7	model_no	character varying(100)	NULL	-	-
8	serial_no	character varying(100)	NULL	-	-
9	specs	jsonb	NULL	-	-
10	status	character varying(20)	NULL	'active'	-
11	notes	text	NULL	-	-
12	purchase_date	timestamp without time zone	NULL	-	-
13	warranty_end	timestamp without time zone	NULL	-	-
14	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
15	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
16	quantity	integer	NULL	1	-
17	image_url	text	NULL	-	-

Table Rules & Indexes: [Index lab_items_lab_id_item_number_key]: USING btree (lab_id, item_number)

TABLE: LAB_MAINTENANCE_HISTORY

11 Columns | PK: (id) | #39 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> labs.id] [FK -> users.id] [UNIQUE]
2	lab_id	uuid	NOT NULL	-	[FK -> labs.id]
3	action	character varying(20)	NOT NULL	-	-
4	reason	text	NULL	-	-
5	previous_status	character varying(20)	NULL	-	-
6	new_status	character varying(20)	NOT NULL	-	-
7	started_at	timestamp without time zone	NULL	-	-
8	ended_at	timestamp without time zone	NULL	-	-
9	expected_end_date	timestamp without time zone	NULL	-	-
10	performed_by_id	uuid	NULL	-	[FK -> users.id]
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: LAB_MATERIAL_USAGE

9 Columns | PK: (id) | #40 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> assignments.id] [FK -> lab_materials.id CASCADE] [FK -> users.id] [FK -> users.id] [UNIQUE]
2	material_id	uuid	NOT NULL	-	[FK -> lab_materials.id CASCADE]
3	assignment_id	uuid	NULL	-	[FK -> assignments.id]
4	student_id	uuid	NOT NULL	-	[FK -> users.id]
5	quantity_used	numeric	NOT NULL	-	-
6	usage_date	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
7	recorded_by	uuid	NOT NULL	-	[FK -> users.id]
8	remarks	text	NULL	-	-
9	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: LAB_MATERIALS

10 Columns | PK: (id) | #41 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> labs.id CASCADE] [UNIQUE]
2	lab_id	uuid	NOT NULL	-	[FK -> labs.id CASCADE]
3	name	character varying(100)	NOT NULL	-	-
4	name_hindi	character varying(100)	NULL	-	-
5	description	text	NULL	-	-
6	unit	enum (MaterialUnit)	NOT NULL	-	-
7	cost_per_unit	numeric	NULL	-	-
8	current_stock	numeric	NULL	0	-
9	minimum_stock	numeric	NULL	0	-

TABLE: LAB_MATERIALS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
10	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: LABS

14 Columns | PK: (id) | #42 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> schools.id CASCADE] [FK -> subjects.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	name	character varying(100)	NOT NULL	-	-
4	name_hindi	character varying(100)	NULL	-	-
5	room_number	character varying(20)	NULL	-	-
6	capacity	integer	NULL	30	-
7	subject_id	uuid	NULL	-	[FK -> subjects.id]
8	equipment_list	jsonb	NULL	-	-
9	incharge_id	uuid	NULL	-	[FK -> users.id]
10	status	character varying(20)	NULL	'active'	-
11	maintenance_reason	text	NULL	-	-
12	maintenance_start_date	timestamp without time zone	NULL	-	-
13	maintenance_end_date	timestamp without time zone	NULL	-	-
14	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: LAPTOP_ISSUANCES

20 Columns | PK: (id) | #43 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> users.id] [FK -> lab_items.id] [FK -> users.id] [FK -> schools.id] [UNIQUE]
2	laptop_id	uuid	NOT NULL	-	[FK -> lab_items.id]
3	issued_to_id	uuid	NOT NULL	-	[FK -> users.id]
4	issued_by_id	uuid	NOT NULL	-	[FK -> users.id]
5	voucher_number	character varying(30)	NOT NULL	-	[UNIQUE]
6	purpose	text	NULL	-	-
7	issued_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
8	expected_return_date	date	NULL	-	-
9	returned_at	timestamp without time zone	NULL	-	-
10	received_by_id	uuid	NULL	-	[FK -> users.id]
11	condition_on_issue	character varying(50)	NULL	'good'	-
12	condition_on_return	character varying(50)	NULL	-	-
13	component_status	jsonb	NULL	'{}'	-
14	component_status_on_return	jsonb	NULL	-	-

TABLE: LAPTOP_ISSUANCES (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
15	remarks	text	NULL	-	-
16	return_remarks	text	NULL	-	-
17	status	enum (issuance_status)	NOT NULL	'issued'	-
18	school_id	uuid	NOT NULL	-	[FK -> schools.id]
19	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
20	updated_at	timestamp without time zone	NOT NULL	-	-

Table Rules & Indexes: [Index laptop_issuances_issued_to_id_idx]: USING btree (issued_to_id) | [Index laptop_issuances_laptop_id_idx]: USING btree (laptop_id) | [Index laptop_issuances_school_id_idx]: USING btree (school_id)

TABLE: LECTURE_ATTENDANCE

7 Columns | PK: (id) | #44 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lecture_sessions.id] [FK -> users.id] [UNIQUE]
2	lecture_session_id	uuid	NOT NULL	-	[FK -> lecture_sessions.id]
3	student_id	uuid	NOT NULL	-	[FK -> users.id]
4	status	enum (AttendanceStatus)	NOT NULL	'present'::"AttendanceSta...	-
5	joined_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
6	left_at	timestamp without time zone	NULL	-	-
7	engagement_score	integer	NULL	-	-

Table Rules & Indexes: [Index lecture_attendance_lecture_session_id_student_id_key]: USING btree (lecture_session_id, student_id) | [Index lecture_attendance_student_id_idx]: USING btree (student_id)

TABLE: LECTURE_PLANS

20 Columns | PK: (id) | #45 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> classes.id] [FK -> users.id] [FK -> schools.id] [FK -> subjects.id] [FK -> timetable_slots.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]
3	instructor_id	uuid	NOT NULL	-	[FK -> users.id]
4	subject_id	uuid	NOT NULL	-	[FK -> subjects.id]
5	class_id	uuid	NOT NULL	-	[FK -> classes.id]
6	timetable_slot_id	uuid	NULL	-	[FK -> timetable_slots.id]
7	academic_year_id	uuid	NOT NULL	-	[FK -> academic_years.id]
8	title	character varying(500)	NOT NULL	-	-
9	title_hindi	character varying(500)	NULL	-	-
10	description	text	NULL	-	-
11	lecture_number	integer	NOT NULL	-	-
12	scheduled_date	date	NOT NULL	-	-
13	scheduled_duration	integer	NOT NULL	40	-
14	lecture_type	enum (lecture_type)	NOT NULL	'theory'	-

TABLE: LECTURE_PLANS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
15	status	enum (lecture_plan_status)	NOT NULL	'planned'	-
16	notes	text	NULL	-	-
17	notes_hindi	text	NULL	-	-
18	homework_description	text	NULL	-	-
19	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
20	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index lecture_plans_class_id_subject_id_idx]: USING btree (class_id, subject_id) | [Index lecture_plans_instructor_id_idx]: USING btree (instructor_id) | [Index lecture_plans_scheduled_date_idx]: USING btree (scheduled_date)

TABLE: LECTURE_POLL_RESPONSES

5 Columns | PK: (id) | #46 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lecture_polls.id] [FK -> users.id] [UNIQUE]
2	poll_id	uuid	NOT NULL	-	[FK -> lecture_polls.id]
3	student_id	uuid	NOT NULL	-	[FK -> users.id]
4	selected_option	integer	NOT NULL	-	-
5	answered_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index lecture_poll_responses_poll_id_student_id_key]: USING btree (poll_id, student_id)

TABLE: LECTURE_POLLS

7 Columns | PK: (id) | #47 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lecture_sessions.id] [UNIQUE]
2	lecture_session_id	uuid	NOT NULL	-	[FK -> lecture_sessions.id]
3	question	text	NOT NULL	-	-
4	options	jsonb	NOT NULL	-	-
5	correct_option	integer	NULL	-	-
6	is_active	boolean	NOT NULL	true	-
7	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index lecture_polls_lecture_session_id_idx]: USING btree (lecture_session_id)

TABLE: LECTURE_RESOURCES

11 Columns | PK: (id) | #48 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lecture_plans.id] [FK -> lecture_sessions.id] [FK -> users.id] [UNIQUE]
2	lecture_plan_id	uuid	NULL	-	[FK -> lecture_plans.id]
3	lecture_session_id	uuid	NULL	-	[FK -> lecture_sessions.id]
4	title	character varying(500)	NOT NULL	-	-
5	type	enum (resource_type)	NOT NULL	'pdf'	-

TABLE: LECTURE_RESOURCES (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
6	**url**	character varying(1000)	NOT NULL	-	-
7	**file_size**	bigint	NULL	-	-
8	**mime_type**	character varying(100)	NULL	-	-
9	**uploaded_by_id**	uuid	NOT NULL	-	[FK -> users.id]
10	**sequence_order**	integer	NOT NULL	0	-
11	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index lecture_resources_lecture_plan_id_idx]: USING btree (lecture_plan_id) | [Index lecture_resources_lecture_session_id_idx]: USING btree (lecture_session_id)

TABLE: LECTURE_SESSIONS

10 Columns | PK: (id) | #49 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lecture_plans.id] [UNIQUE]
2	**lecture_plan_id**	uuid	NOT NULL	-	[FK -> lecture_plans.id]
3	**started_at**	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
4	**ended_at**	timestamp without time zone	NULL	-	-
5	**actual_duration**	integer	NULL	-	-
6	**attendance_count**	integer	NOT NULL	0	-
7	**topics_covered**	jsonb	NULL	-	-
8	**instructor_remarks**	text	NULL	-	-
9	**recording_url**	character varying(1000)	NULL	-	-
10	**status**	enum (lecture_session_status)	NOT NULL	'active'	-

Table Rules & Indexes: [Index lecture_sessions_lecture_plan_id_idx]: USING btree (lecture_plan_id) | [Index lecture_sessions_started_at_idx]: USING btree (started_at)

TABLE: MEETING_PARTICIPANTS

11 Columns | PK: (id) | #50 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> meetings.id CASCADE] [FK -> users.id] [UNIQUE]
2	**session_id**	uuid	NOT NULL	-	[FK -> meetings.id CASCADE]
3	**user_id**	uuid	NOT NULL	-	[FK -> users.id]
4	**role**	character varying(50)	NOT NULL	'student'	-
5	**status**	enum (ParticipantStatus)	NOT NULL	'waiting'::"ParticipantSt...	-
6	**joined_waiting_at**	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
7	**admitted_at**	timestamp without time zone	NULL	-	-
8	**left_at**	timestamp without time zone	NULL	-	-
9	**socket_id**	character varying(255)	NULL	-	-
10	**is_video_enabled**	boolean	NOT NULL	false	-
11	**is_audio_enabled**	boolean	NOT NULL	false	-

Table Rules & Indexes: [Index idx_meeting_participants_session]: USING btree (session_id) | [Index idx_meeting_participants_status]: USING btree (status) | [Index idx_meeting_participants_user]: USING btree (user_id)

TABLE: MEETING_QUESTIONS

12 Columns | PK: (id) | #51 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> assignments.id] [FK -> users.id] [FK -> subjects.id CASCADE] [UNIQUE]
2	**subject_id**	uuid	NOT NULL	-	[FK -> subjects.id CASCADE]
3	**assignment_id**	uuid	NULL	-	[FK -> assignments.id]
4	**question**	text	NOT NULL	-	-
5	**question_hindi**	text	NULL	-	-
6	**expected_answer**	text	NULL	-	-
7	**expected_answer_hindi**	text	NULL	-	-
8	**difficulty**	enum (DifficultyLevel)	NULL	'medium'::"DifficultyLevel"	-
9	**marks**	integer	NULL	2	-
10	**topic_tags**	ARRAY	NULL	-	-
11	**created_by**	uuid	NOT NULL	-	[FK -> users.id]
12	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: MEETINGS

32 Columns | PK: (id) | #52 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> schools.id CASCADE] [FK -> submissions.id CASCADE] [FK -> classes.id CASCADE] [FK -> student_groups.id CASCADE] [FK -> users.id] [UNIQUE]
2	**school_id**	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	**title**	character varying(255)	NOT NULL	-	-
4	**type**	character varying(50)	NOT NULL	'instant'	-
5	**target_class_id**	uuid	NULL	-	[FK -> classes.id CASCADE]
6	**target_group_id**	uuid	NULL	-	[FK -> student_groups.id CASCADE]
7	**target_student_id**	uuid	NULL	-	[FK -> users.id]
8	**submission_id**	uuid	NULL	-	[FK -> submissions.id CASCADE]
9	**host_id**	uuid	NOT NULL	-	[FK -> users.id]
10	**scheduled_at**	timestamp without time zone	NULL	-	-
11	**duration_minutes**	integer	NULL	10	-
12	**actual_start_time**	timestamp without time zone	NULL	-	-
13	**actual_end_time**	timestamp without time zone	NULL	-	-
14	**mode**	enum (MeetingMode)	NULL	'online'::"MeetingMode"	-
15	**meeting_link**	text	NULL	-	-
16	**recording_url**	text	NULL	-	-
17	**questions_asked**	jsonb	NULL	-	-
18	**student_responses**	jsonb	NULL	-	-
19	**marks_obtained**	numeric	NULL	-	-

TABLE: MEETINGS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
20	max_marks	numeric	NULL	-	-
21	performance_rating	integer	NULL	-	-
22	examiner_remarks	text	NULL	-	-
23	examiner_remarks_hindi	text	NULL	-	-
24	improvement_suggestions	text	NULL	-	-
25	status	enum (MeetingStatus)	NULL	'scheduled'::"MeetingStatus"	-
26	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
27	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
28	scheduled_end_time	timestamp without time zone	NULL	-	-
29	auto_start	boolean	NULL	true	-
30	recording_file_path	text	NULL	-	-
31	recording_size_bytes	integer	NULL	-	-
32	recording_duration_seconds	integer	NULL	-	-

Table Rules & Indexes: [Index idx_meetings_auto_start]: USING btree (status, auto_start, scheduled_at)

TABLE: NOTIFICATION_TEMPLATES

11 Columns | PK: (id) | #53 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id CASCADE] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	name	character varying(100)	NOT NULL	-	-
4	subject	character varying(255)	NULL	-	-
5	subject_hindi	character varying(255)	NULL	-	-
6	body_template	text	NOT NULL	-	-
7	body_template_hindi	text	NULL	-	-
8	trigger_event	character varying(100)	NULL	-	-
9	channels	ARRAY	NULL	-	-
10	is_active	boolean	NULL	true	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: NOTIFICATIONS

17 Columns | PK: (id) | #54 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id CASCADE] [UNIQUE]
2	user_id	uuid	NOT NULL	-	[FK -> users.id CASCADE]
3	title	character varying(255)	NOT NULL	-	-
4	title_hindi	character varying(255)	NULL	-	-

TABLE: NOTIFICATIONS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
5	message	text	NOT NULL	-	-
6	message_hindi	text	NULL	-	-
7	type	character varying(50)	NULL	-	-
8	reference_type	character varying(50)	NULL	-	-
9	reference_id	uuid	NULL	-	-
10	is_read	boolean	NULL	false	-
11	read_at	timestamp without time zone	NULL	-	-
12	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
13	template_id	uuid	NULL	-	-
14	action_url	text	NULL	-	-
15	channel	text	NULL	'in_app'	-
16	scheduled_for	timestamp without time zone	NULL	-	-
17	sent_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Index idx_notifications_user]: USING btree (user_id)

TABLE: PROCUREMENT_COMMITTEE

6 Columns | PK: (id) | #55 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> procurement_requests.id] [FK -> users.id] [UNIQUE]
2	request_id	uuid	NOT NULL	-	[FK -> procurement_requests.id]
3	user_id	uuid	NOT NULL	-	[FK -> users.id]
4	role	character varying(50)	NULL	'member'	-
5	designation	character varying(100)	NULL	-	-
6	added_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index procurement_committee_request_id_idx]: USING btree (request_id) | [Index procurement_committee_request_id_user_id_key]: USING btree (request_id, user_id) | [Index procurement_committee_user_id_idx]: USING btree (user_id)

TABLE: PROCUREMENT_ITEMS

17 Columns | PK: (id) | #56 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> vendors.id] [FK -> lab_items.id] [FK -> users.id] [FK -> procurement_requests.id] [UNIQUE]
2	request_id	uuid	NOT NULL	-	[FK -> procurement_requests.id]
3	item_name	character varying(200)	NOT NULL	-	-
4	description	text	NULL	-	-
5	specifications	text	NULL	-	-
6	quantity	integer	NOT NULL	1	-
7	unit	character varying(50)	NULL	'pcs'	-
8	estimated_unit_price	numeric	NULL	-	-

TABLE: PROCUREMENT_ITEMS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
9	approved_unit_price	numeric	NULL	-	-
10	approved_vendor_id	uuid	NULL	-	[FK -> vendors.id]
11	is_received	boolean	NOT NULL	false	-
12	received_qty	integer	NOT NULL	0	-
13	serial_numbers	jsonb	NULL	-	-
14	received_at	timestamp without time zone	NULL	-	-
15	received_by_id	uuid	NULL	-	[FK -> users.id]
16	inventory_item_id	uuid	NULL	-	[FK -> lab_items.id]
17	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index procurement_items_is_received_idx]: USING btree (is_received) | [Index procurement_items_request_id_idx]: USING btree (request_id)

TABLE: PROCUREMENT_REQUESTS

37 Columns | PK: (id) | #57 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> users.id] [FK -> users.id] [FK -> schools.id] [UNIQUE]
2	title	character varying(200)	NOT NULL	-	-
3	description	text	NULL	-	-
4	purpose	text	NULL	-	-
5	department	character varying(100)	NULL	-	-
6	budget_code	character varying(50)	NULL	-	-
7	purchase_letter_url	character varying(500)	NULL	-	-
8	purchase_letter_name	character varying(255)	NULL	-	-
9	letter_content	text	NULL	-	-
10	current_step	integer	NULL	1	-
11	estimated_total	numeric	NULL	-	-
12	approved_total	numeric	NULL	-	-
13	status	enum (procurement_status)	NOT NULL	'draft'	-
14	created_by_id	uuid	NOT NULL	-	[FK -> users.id]
15	approved_by_id	uuid	NULL	-	[FK -> users.id]
16	approved_at	timestamp without time zone	NULL	-	-
17	ordered_at	timestamp without time zone	NULL	-	-
18	po_number	character varying(100)	NULL	-	-
19	po_url	character varying(500)	NULL	-	-
20	bill_number	character varying(100)	NULL	-	-
21	bill_date	date	NULL	-	-
22	bill_amount	numeric	NULL	-	-
23	bill_url	character varying(500)	NULL	-	-

TABLE: PROCUREMENT_REQUESTS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
24	payment_method	character varying(50)	NULL	-	-
25	cheque_number	character varying(50)	NULL	-	-
26	cheque_url	character varying(500)	NULL	-	-
27	payment_date	date	NULL	-	-
28	payment_reference	character varying(100)	NULL	-	-
29	received_at	timestamp without time zone	NULL	-	-
30	received_by_id	uuid	NULL	-	[FK -> users.id]
31	receiving_video_url	character varying(500)	NULL	-	-
32	receiving_notes	text	NULL	-	-
33	memo_vide_no	character varying(100)	NULL	-	-
34	proceedings_date	date	NULL	-	-
35	school_id	uuid	NOT NULL	-	[FK -> schools.id]
36	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
37	updated_at	timestamp without time zone	NOT NULL	-	-

Table Rules & Indexes: [Index procurement_requests_school_id_idx]: USING btree (school_id) | [Index procurement_requests_status_idx]: USING btree (status)

TABLE: QUERY_LOGS

12 Columns | PK: (id) | #58 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	query	text	NOT NULL	-	-
3	params	text	NULL	-	-
4	duration	integer	NULL	-	-
5	error	text	NULL	-	-
6	success	boolean	NOT NULL	true	-
7	model	character varying(100)	NULL	-	-
8	action	character varying(50)	NULL	-	-
9	user_id	uuid	NULL	-	-
10	user_email	character varying(255)	NULL	-	-
11	ip_address	character varying(45)	NULL	-	-
12	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index query_logs_created_at_idx]: USING btree (created_at) | [Index query_logs_model_idx]: USING btree (model) | [Index query_logs_success_idx]: USING btree (success)

TABLE: QUESTION_BANK

19 Columns | PK: (id) | #59 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]

TABLE: QUESTION_BANK (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
2	topic	character varying(255)	NOT NULL	-	-
3	exam_type	character varying(100)	NOT NULL	-	-
4	exam_year	character varying(50)	NULL	-	-
5	difficulty	character varying(50)	NULL	'medium'	-
6	question	text	NOT NULL	-	-
7	options	jsonb	NOT NULL	'[]'	-
8	correct_option	character varying(10)	NOT NULL	-	-
9	explanation	text	NULL	-	-
10	points	integer	NOT NULL	1	-
11	source	character varying(100)	NULL	'question_bank'	-
12	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
13	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
14	diagram_url	text	NULL	-	-
15	diagram_svg	text	NULL	-	-
16	question_type	character varying(50)	NULL	'mcq'	-
17	exam_occurrences	jsonb	NULL	'[]'	-
18	is_mock	boolean	NULL	false	-
19	web_source	text	NULL	'Official Paper'	-

Table Rules & Indexes: [Index idx_question_bank_exam_type]: USING btree (exam_type) | [Index idx_question_bank_exam_year]: USING btree (exam_year) | [Index idx_question_bank_lookup]: USING btree (exam_type, exam_year, topic)

TABLE: QUIZ_ASSIGNMENTS

8 Columns | PK: (id) | #60 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id CASCADE] [FK -> classes.id CASCADE] [FK -> student_groups.id CASCADE] [FK -> quizzes.id CASCADE] [FK -> users.id CASCADE] [UNIQUE]
2	quiz_id	uuid	NOT NULL	-	[FK -> quizzes.id CASCADE]
3	target_type	character varying(50)	NOT NULL	-	-
4	target_class_id	uuid	NULL	-	[FK -> classes.id CASCADE]
5	target_group_id	uuid	NULL	-	[FK -> student_groups.id CASCADE]
6	target_student_id	uuid	NULL	-	[FK -> users.id CASCADE]
7	assigned_by_id	uuid	NOT NULL	-	[FK -> users.id CASCADE]
8	assigned_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_quiz_assignments_class_id]: USING btree (target_class_id) | [Index idx_quiz_assignments_group_id]: USING btree (target_group_id) | [Index idx_quiz_assignments_quiz_id]: USING btree (quiz_id)

TABLE: QUIZ_SUBMISSIONS

12 Columns | PK: (id) | #61 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> quizzes.id CASCADE] [FK -> users.id CASCADE] [UNIQUE]

TABLE: QUIZ_SUBMISSIONS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
2	quiz_id	uuid	NOT NULL	-	[FK -> quizzes.id CASCADE]
3	user_id	uuid	NOT NULL	-	[FK -> users.id CASCADE]
4	score	double precision	NOT NULL	0	-
5	max_score	double precision	NOT NULL	0	-
6	percentage	double precision	NOT NULL	0	-
7	correct_answers	integer	NOT NULL	0	-
8	total_questions	integer	NOT NULL	0	-
9	time_taken_seconds	integer	NOT NULL	0	-
10	answers	jsonb	NOT NULL	'[]'	-
11	is_timed_out	boolean	NOT NULL	false	-
12	submitted_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_quiz_submissions_quiz_id]: USING btree (quiz_id) | [Index idx_quiz_submissions_user_id]: USING btree (user_id)

TABLE: QUIZZES

15 Columns | PK: (id) | #62 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	code	character varying(10)	NOT NULL	-	[UNIQUE]
3	title	character varying(255)	NOT NULL	-	-
4	description	text	NULL	-	-
5	keywords	text	NULL	-	-
6	difficulty	character varying(50)	NULL	'medium'	-
7	total_questions	integer	NOT NULL	5	-
8	time_limit_minutes	integer	NOT NULL	10	-
9	questions	jsonb	NOT NULL	'[]'	-
10	status	character varying(50)	NOT NULL	'published'	-
11	created_by_id	uuid	NULL	-	-
12	school_id	uuid	NULL	-	-
13	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
14	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
15	max_attempts	integer	NOT NULL	1	-

Table Rules & Indexes: [Index idx_quizzes_code]: USING btree (code) | [Index idx_quizzes_created_by]: USING btree (created_by_id) | [Index idx_quizzes_status]: USING btree (status)

TABLE: QUOTATION_ITEMS

7 Columns | PK: (id) | #63 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> procurement_items.id] [FK -> vendor_quotations.id] [UNIQUE]

TABLE: QUOTATION_ITEMS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
2	quotation_id	uuid	NOT NULL	-	[FK -> vendor_quotations.id]
3	procurement_item_id	uuid	NOT NULL	-	[FK -> procurement_items.id]
4	unit_price	numeric	NOT NULL	-	-
5	quantity	integer	NULL	-	-
6	total_price	numeric	NULL	-	-
7	remarks	text	NULL	-	-

Table Rules & Indexes: [Index quotation_items_procurement_item_id_idx]: USING btree (procurement_item_id) | [Index quotation_items_quotation_id_idx]: USING btree (quotation_id)

TABLE: REPORT_TEMPLATES

10 Columns | PK: (id) | #64 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id CASCADE] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	name	character varying(100)	NOT NULL	-	-
4	type	enum (ReportType)	NOT NULL	-	-
5	template_content	text	NULL	-	-
6	header_html	text	NULL	-	-
7	footer_html	text	NULL	-	-
8	styles	text	NULL	-	-
9	is_default	boolean	NULL	false	-
10	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: SCHOOL_CALENDAR

14 Columns | PK: (id) | #65 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> users.id] [FK -> schools.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]
3	academic_year_id	uuid	NOT NULL	-	[FK -> academic_years.id]
4	date	date	NOT NULL	-	-
5	title	character varying(255)	NOT NULL	-	-
6	title_hindi	character varying(255)	NULL	-	-
7	type	enum (calendar_event_type)	NOT NULL	'gazetted_holiday'	-
8	is_holiday	boolean	NOT NULL	true	-
9	start_time	character varying(20)	NULL	-	-
10	end_time	character varying(20)	NULL	-	-
11	description	text	NULL	-	-
12	source	enum (calendar_source)	NOT NULL	'admin_custom'	-

TABLE: SCHOOL_CALENDAR (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
13	created_by_id	uuid	NULL	-	[FK -> users.id]
14	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index school_calendar_date_idx]: USING btree (date) | [Index school_calendar_school_id_academic_year_id_idx]: USING btree (school_id, academic_year_id) | [Index school_calendar_school_id_date_idx]: USING btree (school_id, date)

TABLE: SCHOOLS

20 Columns | PK: (id) | #66 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	name	character varying(255)	NOT NULL	-	-
3	name_hindi	character varying(255)	NULL	-	-
4	name_regional	character varying(255)	NULL	-	-
5	code	character varying(50)	NOT NULL	-	[UNIQUE]
6	address	text	NULL	-	-
7	state	character varying(100)	NULL	-	-
8	district	character varying(100)	NULL	-	-
9	board_affiliation	character varying(100)	NULL	-	-
10	primary_language	character varying(10)	NULL	'en'	-
11	secondary_languages	ARRAY	NULL	-	-
12	academic_year_start	integer	NULL	4	-
13	logo_url	text	NULL	-	-
14	letterhead_url	character varying(500)	NULL	-	-
15	email	character varying(255)	NULL	-	-
16	phone1	character varying(20)	NULL	-	-
17	phone2	character varying(20)	NULL	-	-
18	pin_code	character varying(10)	NULL	-	-
19	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
20	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index schools_code_key]: USING btree (code)

TABLE: SITE_UPDATES

6 Columns | PK: (id) | #67 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	version	character varying(20)	NOT NULL	-	-
3	description	text	NOT NULL	-	-
4	changes	text	NULL	-	-
5	updated_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

TABLE: SITE_UPDATES (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
6	updated_by	character varying(100)	NULL	-	-

TABLE: STUDENT_FEES

11 Columns | PK: (id) | #68 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> fee_structures.id] [UNIQUE]
2	student_id	uuid	NOT NULL	-	-
3	fee_structure_id	uuid	NOT NULL	-	[FK -> fee_structures.id]
4	academic_year_id	uuid	NOT NULL	-	[FK -> academic_years.id]
5	base_amount	numeric	NOT NULL	-	-
6	concession_amount	numeric	NULL	0	-
7	concession_reason	text	NULL	-	-
8	final_amount	numeric	NOT NULL	-	-
9	status	enum (FeeStatus)	NULL	'pending'::"FeeStatus"	-
10	due_date	timestamp without time zone	NULL	-	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: STUDENT_GROUPS

8 Columns | PK: (id) | #69 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> lab_items.id] [FK -> classes.id CASCADE] [FK -> users.id] [UNIQUE]
2	class_id	uuid	NOT NULL	-	[FK -> classes.id CASCADE]
3	name	character varying(100)	NOT NULL	-	-
4	name_hindi	character varying(100)	NULL	-	-
5	description	text	NULL	-	-
6	created_by	uuid	NOT NULL	-	[FK -> users.id]
7	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
8	assigned_pc_id	uuid	NULL	-	[FK -> lab_items.id]

Table Rules & Indexes: [Index student_groups_class_id_name_key]: USING btree (class_id, name)

TABLE: STUDENT_TRAINING_PROGRESS

10 Columns | PK: (id) | #70 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> training_modules.id] [FK -> users.id] [UNIQUE]
2	student_id	uuid	NOT NULL	-	[FK -> users.id]
3	module_id	uuid	NOT NULL	-	[FK -> training_modules.id]
4	current_unit_id	uuid	NULL	-	-
5	overall_progress	double precision	NOT NULL	0	-

TABLE: STUDENT_TRAINING_PROGRESS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
6	total_xp	integer	NOT NULL	0	-
7	streak	integer	NOT NULL	0	-
8	last_active_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
9	started_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
10	completed_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Index student_training_progress_student_id_module_id_key]: USING btree (student_id, module_id)

TABLE: STUDENT_UNIT_MASTERY

8 Columns | PK: (id) | #71 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [UNIQUE]
2	student_id	uuid	NOT NULL	-	[FK -> users.id]
3	unit_id	uuid	NOT NULL	-	-
4	mastery_score	double precision	NOT NULL	0	-
5	exercises_done	integer	NOT NULL	0	-
6	status	text	NOT NULL	'locked'	-
7	unlocked_at	timestamp without time zone	NULL	-	-
8	mastered_at	timestamp without time zone	NULL	-	-

Table Rules & Indexes: [Index student_unit_mastery_student_id_unit_id_key]: USING btree (student_id, unit_id)

TABLE: SUBJECTS

10 Columns | PK: (id) | #72 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id CASCADE] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	code	character varying(20)	NOT NULL	-	-
4	name	character varying(100)	NOT NULL	-	-
5	name_hindi	character varying(100)	NULL	-	-
6	name_regional	character varying(100)	NULL	-	-
7	has_lab	boolean	NULL	false	-
8	lab_hours_per_week	integer	NULL	0	-
9	theory_hours_per_week	integer	NULL	0	-
10	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: SUBMISSION_FILES

8 Columns | PK: (id) | #73 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> submissions.id CASCADE] [UNIQUE]

TABLE: SUBMISSION_FILES (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
2	submission_id	uuid	NOT NULL	-	[FK -> submissions.id CASCADE]
3	file_name	character varying(255)	NOT NULL	-	-
4	file_type	character varying(50)	NULL	-	-
5	file_size	integer	NULL	-	-
6	file_url	text	NOT NULL	-	-
7	is_output	boolean	NULL	false	-
8	uploaded_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: SUBMISSION_REVISIONS

8 Columns | PK: (id) | #74 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> submissions.id CASCADE] [UNIQUE]
2	submission_id	uuid	NOT NULL	-	[FK -> submissions.id CASCADE]
3	revision_number	integer	NOT NULL	-	-
4	code_content	text	NULL	-	-
5	output_content	text	NULL	-	-
6	attachments	jsonb	NULL	-	-
7	revision_note	text	NULL	-	-
8	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: SUBMISSIONS

18 Columns | PK: (id) | #75 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> assignments.id CASCADE] [FK -> student_groups.id] [FK -> users.id CASCADE] [UNIQUE]
2	assignment_id	uuid	NOT NULL	-	[FK -> assignments.id CASCADE]
3	student_id	uuid	NOT NULL	-	[FK -> users.id CASCADE]
4	group_id	uuid	NULL	-	[FK -> student_groups.id]
5	code_content	text	NULL	-	-
6	output_content	text	NULL	-	-
7	observations	text	NULL	-	-
8	observations_hindi	text	NULL	-	-
9	conclusion	text	NULL	-	-
10	conclusion_hindi	text	NULL	-	-
11	attachments	jsonb	NULL	-	-
12	output_screenshot_url	text	NULL	-	-
13	submission_number	integer	NULL	1	-
14	is_late	boolean	NULL	false	-

TABLE: SUBMISSIONS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
15	late_days	integer	NULL	0	-
16	status	enum (SubmissionStatus)	NULL	'submitted'::"SubmissionS...	-
17	submitted_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
18	last_modified	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_submissions_assignment]: USING btree (assignment_id) | [Index idx_submissions_student]: USING btree (student_id) | [Index submissions_assignment_id_idx]: USING btree (assignment_id)

TABLE: SYSTEM_SETTINGS

5 Columns | PK: (id) | #76 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	key	character varying(100)	NOT NULL	-	[UNIQUE]
3	value	jsonb	NOT NULL	-	-
4	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
5	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_system_settings_key]: USING btree (key)

TABLE: TICKET_COMMENTS

5 Columns | PK: (id) | #77 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> tickets.id] [FK -> users.id] [UNIQUE]
2	ticket_id	uuid	NOT NULL	-	[FK -> tickets.id]
3	user_id	uuid	NOT NULL	-	[FK -> users.id]
4	content	text	NOT NULL	-	-
5	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

TABLE: TICKET_ISSUE_TYPES

7 Columns | PK: (id) | #78 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	category	enum (ticket_category)	NOT NULL	-	-
3	name	character varying(100)	NOT NULL	-	-
4	description	text	NULL	-	-
5	display_order	integer	NOT NULL	0	-
6	is_active	boolean	NOT NULL	true	-
7	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index ticket_issue_types_category_idx]: USING btree (category)

TABLE: TICKETS

17 Columns | PK: (id) | #79 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> users.id] [FK -> ticket_issue_types.id] [FK -> lab_items.id] [FK -> labs.id] [FK -> users.id] [UNIQUE]
2	ticket_number	character varying(20)	NOT NULL	-	[UNIQUE]
3	title	character varying(200)	NOT NULL	-	-
4	description	text	NOT NULL	-	-
5	category	enum (ticket_category)	NOT NULL	'other'	-
6	priority	enum (ticket_priority)	NOT NULL	'medium'	-
7	status	enum (ticket_status)	NOT NULL	'open'	-
8	item_id	uuid	NULL	-	[FK -> lab_items.id]
9	lab_id	uuid	NULL	-	[FK -> labs.id]
10	issue_type_id	uuid	NULL	-	[FK -> ticket_issue_types.id]
11	created_by_id	uuid	NOT NULL	-	[FK -> users.id]
12	assigned_to_id	uuid	NULL	-	[FK -> users.id]
13	resolved_by_id	uuid	NULL	-	[FK -> users.id]
14	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
15	updated_at	timestamp without time zone	NOT NULL	-	-
16	resolved_at	timestamp without time zone	NULL	-	-
17	resolution_notes	text	NULL	-	-

Table Rules & Indexes: [Index tickets_created_by_id_idx]: USING btree (created_by_id) | [Index tickets_lab_id_idx]: USING btree (lab_id) | [Index tickets_status_idx]: USING btree (status)

TABLE: TIMETABLE_SLOTS

11 Columns | PK: (id) | #80 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> subjects.id] [FK -> timetables.id] [UNIQUE]
2	timetable_id	uuid	NOT NULL	-	[FK -> timetables.id]
3	day_of_week	enum (DayOfWeek)	NOT NULL	-	-
4	period_number	integer	NOT NULL	-	-
5	start_time	character varying(5)	NOT NULL	-	-
6	end_time	character varying(5)	NOT NULL	-	-
7	subject_id	uuid	NULL	-	[FK -> subjects.id]
8	instructor_id	uuid	NULL	-	[FK -> users.id]
9	room_number	character varying(50)	NULL	-	-
10	slot_type	enum (slot_type)	NOT NULL	'lecture'	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index timetable_slots_instructor_id_idx]: USING btree (instructor_id) | [Index timetable_slots_subject_id_idx]: USING btree (subject_id) | [Index timetable_slots_timetable_id_day_of_week_period_number_key]: USING btree (timetable_id, day_of_week, period_number)

TABLE: TIMETABLES

10 Columns | PK: (id) | #81 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [FK -> classes.id] [FK -> schools.id] [UNIQUE]
2	**school_id**	uuid	NOT NULL	-	[FK -> schools.id]
3	**class_id**	uuid	NOT NULL	-	[FK -> classes.id]
4	**academic_year_id**	uuid	NOT NULL	-	[FK -> academic_years.id]
5	**name**	character varying(255)	NOT NULL	-	-
6	**effective_from**	date	NOT NULL	-	-
7	**effective_to**	date	NULL	-	-
8	**is_active**	boolean	NOT NULL	true	-
9	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
10	**updated_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index timetables_academic_year_id_idx]: USING btree (academic_year_id) | [Index timetables_class_id_academic_year_id_key]: USING btree (class_id, academic_year_id) | [Index timetables_class_id_idx]: USING btree (class_id)

TABLE: TRAINING_EXERCISES

18 Columns | PK: (id) | #82 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> training_units.id] [UNIQUE]
2	**unit_id**	uuid	NOT NULL	-	[FK -> training_units.id]
3	**title**	character varying(255)	NOT NULL	-	-
4	**description**	text	NOT NULL	-	-
5	**difficulty**	text	NOT NULL	-	-
6	**scaffold_level**	text	NOT NULL	-	-
7	**exercise_type**	text	NOT NULL	'coding'	-
8	**blooms_level**	text	NULL	-	-
9	**learning_objective**	text	NULL	-	-
10	**is_review_exercise**	boolean	NOT NULL	false	-
11	**reviews_topic_id**	uuid	NULL	-	-
12	**starter_code**	text	NULL	-	-
13	**solution_code**	text	NULL	-	-
14	**test_cases**	jsonb	NULL	-	-
15	**hints**	jsonb	NULL	-	-
16	**time_limit**	integer	NOT NULL	5	-
17	**sequence_order**	integer	NOT NULL	0	-
18	**xp_reward**	integer	NOT NULL	10	-

TABLE: TRAINING_MODULES

15 Columns | PK: (id) | #83 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> academic_years.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	-
3	title	character varying(255)	NOT NULL	-	-
4	title_hindi	character varying(255)	NULL	-	-
5	description	text	NULL	-	-
6	language	character varying(50)	NOT NULL	-	-
7	board_aligned	character varying(50)	NULL	-	-
8	class_level	integer	NULL	-	-
9	total_units	integer	NOT NULL	0	-
10	total_exercises	integer	NOT NULL	0	-
11	is_published	boolean	NOT NULL	false	-
12	pedagogy_config	jsonb	NULL	'{}'	-
13	academic_year_id	uuid	NULL	-	[FK -> academic_years.id]
14	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
15	updated_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

TABLE: TRAINING_UNITS

8 Columns | PK: (id) | #84 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> training_modules.id] [UNIQUE]
2	module_id	uuid	NOT NULL	-	[FK -> training_modules.id]
3	unit_number	integer	NOT NULL	-	-
4	title	character varying(255)	NOT NULL	-	-
5	description	text	NULL	-	-
6	expected_hours	integer	NULL	-	-
7	unlock_threshold	integer	NOT NULL	80	-
8	sequence_order	integer	NOT NULL	0	-

TABLE: TRANSLATIONS

6 Columns | PK: (id) | #85 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [UNIQUE]
2	key	character varying(255)	NOT NULL	-	-
3	language_code	character varying(10)	NOT NULL	-	-
4	value	text	NOT NULL	-	-
5	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
6	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index translations_key_language_code_key]: USING btree (key, language_code) | [Index translations_language_code_idx]: USING btree (language_code)

TABLE: USER_SESSIONS

7 Columns | PK: (id) | #86 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id CASCADE] [UNIQUE]
2	**user_id**	uuid	NOT NULL	-	[FK -> users.id CASCADE]
3	**token**	text	NOT NULL	-	-
4	**ip_address**	character varying(50)	NULL	-	-
5	**user_agent**	text	NULL	-	-
6	**expires_at**	timestamp without time zone	NOT NULL	-	-
7	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

TABLE: USERS

34 Columns | PK: (id) | #87 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	**id**	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id CASCADE] [UNIQUE]
2	**school_id**	uuid	NOT NULL	-	[FK -> schools.id CASCADE]
3	**email**	character varying(255)	NOT NULL	-	[UNIQUE]
4	**phone**	character varying(20)	NULL	-	-
5	**password_hash**	text	NOT NULL	-	-
6	**role**	enum (UserRole)	NOT NULL	-	-
7	**first_name**	character varying(100)	NOT NULL	-	-
8	**first_name_hindi**	character varying(100)	NULL	-	-
9	**first_name_regional**	character varying(100)	NULL	-	-
10	**last_name**	character varying(100)	NOT NULL	-	-
11	**last_name_hindi**	character varying(100)	NULL	-	-
12	**last_name_regional**	character varying(100)	NULL	-	-
13	**admission_number**	character varying(50)	NULL	-	-
14	**employee_id**	character varying(50)	NULL	-	-
15	**profile_image_url**	text	NULL	-	-
16	**preferred_language**	character varying(10)	NULL	'en'	-
17	**is_active**	boolean	NULL	true	-
18	**last_login**	timestamp without time zone	NULL	-	-
19	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
20	**updated_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
21	**student_id**	character varying(50)	NULL	-	-
22	**gender**	character varying(20)	NULL	-	-
23	**reset_pin**	character varying(255)	NULL	-	-
24	**pin_expires_at**	timestamp without time zone	NULL	-	-

TABLE: USERS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
25	storage_quota_mb	integer	NULL	500	-
26	storage_used_bytes	bigint	NULL	0	-
27	security_question	character varying(255)	NULL	-	-
28	security_answer_hash	character varying(255)	NULL	-	-
29	must_change_password	boolean	NULL	false	-
30	parent_name	character varying(100)	NULL	-	-
31	parent_phone	character varying(20)	NULL	-	-
32	parent_email	character varying(255)	NULL	-	-
33	parent_relation	character varying(50)	NULL	'Guardian'	-
34	parent_whatsapp_enabled	boolean	NULL	true	-

Table Rules & Indexes: [Index idx_users_email]: USING btree (email) | [Index idx_users_school]: USING btree (school_id) | [Index idx_users_student_id]: USING btree (student_id)

TABLE: VENDOR_QUOTATIONS

11 Columns | PK: (id) | #88 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> procurement_requests.id] [FK -> vendors.id] [UNIQUE]
2	request_id	uuid	NOT NULL	-	[FK -> procurement_requests.id]
3	vendor_id	uuid	NOT NULL	-	[FK -> vendors.id]
4	quotation_number	character varying(50)	NULL	-	-
5	quotation_date	date	NULL	-	-
6	valid_until	date	NULL	-	-
7	document_url	character varying(500)	NULL	-	-
8	total_amount	numeric	NULL	-	-
9	terms	text	NULL	-	-
10	remarks	text	NULL	-	-
11	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index vendor_quotations_request_id_idx]: USING btree (request_id) | [Index vendor_quotations_vendor_id_idx]: USING btree (vendor_id)

TABLE: VENDORS

11 Columns | PK: (id) | #89 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id] [UNIQUE]
2	name	character varying(200)	NOT NULL	-	-
3	contact_person	character varying(100)	NULL	-	-
4	email	character varying(100)	NULL	-	-
5	phone	character varying(20)	NULL	-	-
6	address	text	NULL	-	-

TABLE: VENDORS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
7	gstin	character varying(20)	NULL	-	-
8	is_local	boolean	NOT NULL	false	-
9	school_id	uuid	NOT NULL	-	[FK -> schools.id]
10	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-
11	updated_at	timestamp without time zone	NOT NULL	-	-

Table Rules & Indexes: [Index vendors_school_id_idx]: USING btree (school_id)

TABLE: WHITEBOARD_FILES

12 Columns | PK: (id) | #90 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> schools.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]
3	owner_id	uuid	NOT NULL	-	[FK -> users.id]
4	title	character varying(255)	NOT NULL	'Untitled Whiteboard'	-
5	description	text	NULL	-	-
6	canvas_data	text	NULL	-	-
7	thumbnail_url	text	NULL	-	-
8	page_count	integer	NULL	1	-
9	is_archived	boolean	NULL	false	-
10	last_opened_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
11	created_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
12	updated_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index whiteboard_files_owner_id_idx]: USING btree (owner_id) | [Index whiteboard_files_school_id_idx]: USING btree (school_id)

TABLE: WHITEBOARD_PARTICIPANTS

7 Columns | PK: (id) | #91 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> whiteboard_sessions.id] [FK -> users.id] [UNIQUE]
2	session_id	uuid	NOT NULL	-	[FK -> whiteboard_sessions.id]
3	user_id	uuid	NOT NULL	-	[FK -> users.id]
4	role	enum (WhiteboardParticipantRole)	NOT NULL	'viewer'::"WhiteboardPart...	-
5	joined_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
6	left_at	timestamp without time zone	NULL	-	-
7	is_active	boolean	NULL	true	-

Table Rules & Indexes: [Index idx_whiteboard_participants_session]: USING btree (session_id) | [Index idx_whiteboard_participants_user]: USING btree (user_id) | [Index unique_participant_per_session]: USING btree (session_id, user_id)

TABLE: WHITEBOARD_RECORDING_SHARES

9 Columns | PK: (id) | #92 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> whiteboard_recordings.id] [FK -> users.id] [FK -> classes.id] [FK -> student_groups.id] [FK -> users.id] [UNIQUE]
2	recording_id	uuid	NOT NULL	-	[FK -> whiteboard_recordings.id]
3	shared_by_id	uuid	NOT NULL	-	[FK -> users.id]
4	target_type	enum (document_share_target_type)	NOT NULL	-	-
5	target_class_id	uuid	NULL	-	[FK -> classes.id]
6	target_group_id	uuid	NULL	-	[FK -> student_groups.id]
7	target_user_id	uuid	NULL	-	[FK -> users.id]
8	message	text	NULL	-	-
9	shared_at	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index whiteboard_recording_shares_recording_id_idx]: USING btree (recording_id) | [Index whiteboard_recording_shares_target_class_id_idx]: USING btree (target_class_id) | [Index whiteboard_recording_shares_target_group_id_idx]: USING btree (target_group_id)

TABLE: WHITEBOARD_RECORDINGS

14 Columns | PK: (id) | #93 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> schools.id] [FK -> users.id] [UNIQUE]
2	user_id	uuid	NOT NULL	-	[FK -> users.id]
3	school_id	uuid	NOT NULL	-	[FK -> schools.id]
4	title	character varying(255)	NOT NULL	-	-
5	description	text	NULL	-	-
6	session_id	character varying(100)	NULL	-	-
7	cloudinary_id	character varying(255)	NOT NULL	-	-
8	cloudinary_url	text	NOT NULL	-	-
9	thumbnail_url	text	NULL	-	-
10	duration	integer	NULL	-	-
11	file_size	integer	NULL	-	-
12	is_public	boolean	NOT NULL	true	-
13	share_token	character varying(64)	NULL	-	[UNIQUE]
14	created_at	timestamp without time zone	NOT NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index whiteboard_recordings_created_at_idx]: USING btree (created_at) | [Index whiteboard_recordings_school_id_idx]: USING btree (school_id) | [Index whiteboard_recordings_share_token_idx]: USING btree (share_token)

TABLE: WHITEBOARD_SESSIONS

17 Columns | PK: (id) | #94 of 94

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
1	id	uuid	NOT NULL	gen_random_uuid()	[PK] [FK -> users.id] [FK -> schools.id] [FK -> classes.id] [FK -> student_groups.id] [UNIQUE]
2	school_id	uuid	NOT NULL	-	[FK -> schools.id]
3	host_id	uuid	NOT NULL	-	[FK -> users.id]
4	title	character varying(255)	NULL	-	-

TABLE: WHITEBOARD_SESSIONS (CONTINUED)

#	COLUMN NAME	DATA TYPE	NULLABLE	DEFAULT	CONSTRAINTS & FOREIGN KEYS
5	**status**	enum (WhiteboardSessionStatus)	NOT NULL	'active'::"WhiteboardSess...	-
6	**target_type**	enum (WhiteboardTargetType)	NULL	-	-
7	**meeting_code**	character varying(8)	NULL	-	[UNIQUE]
8	**target_class_id**	uuid	NULL	-	[FK -> classes.id]
9	**target_group_id**	uuid	NULL	-	[FK -> student_groups.id]
10	**started_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-
11	**ended_at**	timestamp without time zone	NULL	-	-
12	**scheduled_at**	timestamp without time zone	NULL	-	-
13	**duration_minutes**	integer	NULL	-	-
14	**recording_url**	text	NULL	-	-
15	**is_recording**	boolean	NULL	false	-
16	**canvas_data**	text	NULL	-	-
17	**created_at**	timestamp without time zone	NULL	CURRENT_TIMESTAMP	-

Table Rules & Indexes: [Index idx_whiteboard_sessions_host]: USING btree (host_id) | [Index idx_whiteboard_sessions_school]: USING btree (school_id) | [Index idx_whiteboard_sessions_status]: USING btree (status)